A Causal Test of Silent Dissent under Conformity

Peter Máté Lakner and Ayşe Ozdinçer

With Apart Research

Abstract

Agreement among language-model agents can hide different patterns of dependence: an answer may still need the model's own intermediate step, or peer answers may make that step dispensable. Reading the step from a hidden state cannot distinguish these possibilities. We test them by removing a direction associated with the intermediate step, called a bridge, blocking access to peer answers, and combining both edits on the same prompt. On questions selected for bridge dependence alone, three agreeing peers produce contrasting results. With neutral wording, Qwen3.5-4B loses at least 2 log-odds of support for the correct answer only when both routes are removed in 15 of 20 questions: either route can sustain it. Qwen3.6-27B retains dependence on its bridge with peers visible and blocked in 14 of 19. Asking the model to consider peers shifts dependence toward their answers, with mixed results on the 27B model. On 20 new questions frozen in advance, the 27B neutral pattern holds in 17. Moving its bridge toward another word also moves the answer in 19 of 20 neutral questions with peers blocked, under both agreement and disagreement. Controls and validation constrain this interpretation; stricter checks of whether the edit disrupts retrieval weaken claims of non-use. These selected arbitrary lookups establish dependence on tested intermediate signals, not independence in open-ended debate. They show why agreement and readable internal information are insufficient evidence of independent computation.

1 Introduction

Debate and voting combine answers from several model calls to improve reliability (Du et al., 2023; Wang et al., 2022). Their value depends in part on whether agreement reflects separate computations or reliance on the same answers. We study one measurable part of this problem: does an answer still depend on the model's own intermediate step when peer answers are available?

A bridge is the unstated middle step of a question. For example, the Colosseum is in Italy, and Italy starts with I. "Consulted" means that the answer depends on the tested intermediate signal. Reading Italy from a hidden state establishes presence. Removing its direction and measuring the answer tests dependence. It does not establish that the direction represents only Italy.

Our contributions are a same-prompt test of bridge dependence with peers visible and blocked; a peer block checked by a leak test, known-case controls and patching; and evidence that dependence differs by model and wording, with confirmation on new 27B questions. These measurements test a possible source of redundancy in groups, without establishing general agent independence.

Ni and Zou's Silent Dissent (2026) finds that a bridge can rank among the top 100 lens tokens after an agent adopts a wrong majority answer. Their bridge injection is exploratory. We add removal and a peer-side intervention. Our target population is questions where the model demonstrably uses an intermediate step alone, also the population relevant to Silent Dissent's question about an original premise surviving yielding. Different questions and prompts prevent direct comparison of conformity rates. Scripted-peer studies provide related behavioural evidence (Zhu et al., 2025), but repeated answer text can influence replies without a speaker (Hu and Qu, 2026). We therefore use "peer influence" for the prompt manipulation, without inferring a social motive.

Yang et al. (2024) study latent bridge retrieval and use; Yashwanth (2026) studies verbal reports of disagreement. We measure hidden states and dependence. The residual stream is the hidden vector passed between model layers. The Jacobian lens, or J-lens, maps it toward the final representation using averaged

derivatives (Gurnee et al., 2026). J++ filters those derivatives (Ayonrinde and Lindsey, 2026). We use Neuronpedia's fitted lenses to read words and obtain intervention directions. A readable word is evidence about that projection, not the model's complete state.

2 Methods

2.1 Models and task design

The main experiments use Qwen3.5-4B and Qwen3.6-27B with thinking disabled and bfloat16 weights. J++ is primary because the completed runs use it. Earlier J-lens runs provide a development comparison. We record answer probabilities and greedy replies, rather than sampling a population of generated conversations.

The development questions ask for the first letter of a hidden country or city. Number lookups form a further candidate family. Arbitrary lookups reduce overlap between the bridge and answer directions. Natural second hops, such as a country's capital or language, can share information with the bridge and make removal hard to interpret. Peers are scripted inside one user message. They give the correct answer, the same wrong answer, or a hidden answer. Neutral wording says “Give your own answer.” Group-aware wording says “Taking the other agents' answers into account”.

2.2 Selection and bridge interventions

We first select questions for which a bridge intervention can be interpreted. The gate checks both hops and their composition, single-token edited words, solo dependence, and reach: whether the edit disrupts naming the bridge. Probability need is the largest early-window relative fall in correct-answer probability after bridge removal, minus the largest everyday-word fall. The selection threshold is 0.5. Historical development cohorts used need and temporal precedence; we report reach separately rather than claiming every retained question passed the later, stricter gate.

For each retained question, removal projects the hidden vector away from the bridge's lens direction. Edits run from the question's first token to the prompt's end. Windows span 6 layers on the 4B model and 12 on the 27B model, with strides of 3 and 6 layers. Answer onset is the first window where removing the model's answer direction reduces its probability by at least half. Peer-answer positions are excluded from this onset test. Bridge windows must end before onset. Twelve everyday words matched for direction strength estimate a noise floor. Same-kind words, such as other countries, check collateral damage.

To assess whether the intervention reaches the intended information, reach checks whether removal disrupts an added bridge-naming task. Development edits include the added naming positions, which can make reach pass too easily. Confirmation restricts edits to shared positions: median reach in agreeing prompts falls from 0.89–0.95 to 0.13–0.29; across 560 prompts, passes fall from 87% to 47%. Many small-need cases therefore remain mixed. Reach tests retrieval, not whether the next hop reads the bridge. A small removal effect alone does not establish non-use.

To distinguish peer-answer effects from the change in prompt layout, a hidden twin keeps the group layout but replaces peer answers with “?” or “(answer submitted)”. Agreement is compared with this matched layout, not only with the solo prompt used for selection. This limits regression toward weaker effects after selecting unusually strong solo effects. Each wording and peer count has its own hidden twin. The hidden content is not assumed to be neutral.

2.3 Separating bridge and peer dependence

The peer block prevents later positions from reading peer-answer tokens at every layer. It masks access in full-attention layers and zeros the peer tokens' input, update and decay in linear-attention layers. A leak test changes the blocked answers and checks that final logits, the scores before probabilities, are unchanged. Patching is a second method: it replaces hidden vectors at peer positions with those from the matched hidden-answer prompt. Both methods preserve the rest of the prompt.

We run the same prompt with no edit (A), bridge removal (B), peers blocked (C), and both edits (D). Log-odds are $\log[p/(1-p)]$ for the correct-answer probability p. A cell “collapses” if its log-odds fall at least 2

below A. This is a score threshold, not necessarily an answer flip. A−B measures removal with visible peers; C−D measures removal with blocked peers. The decision rule first checks whether C collapses. Otherwise it distinguishes only D, both B and D, neither, and only B collapsing. Appendix B gives the ordered rules.

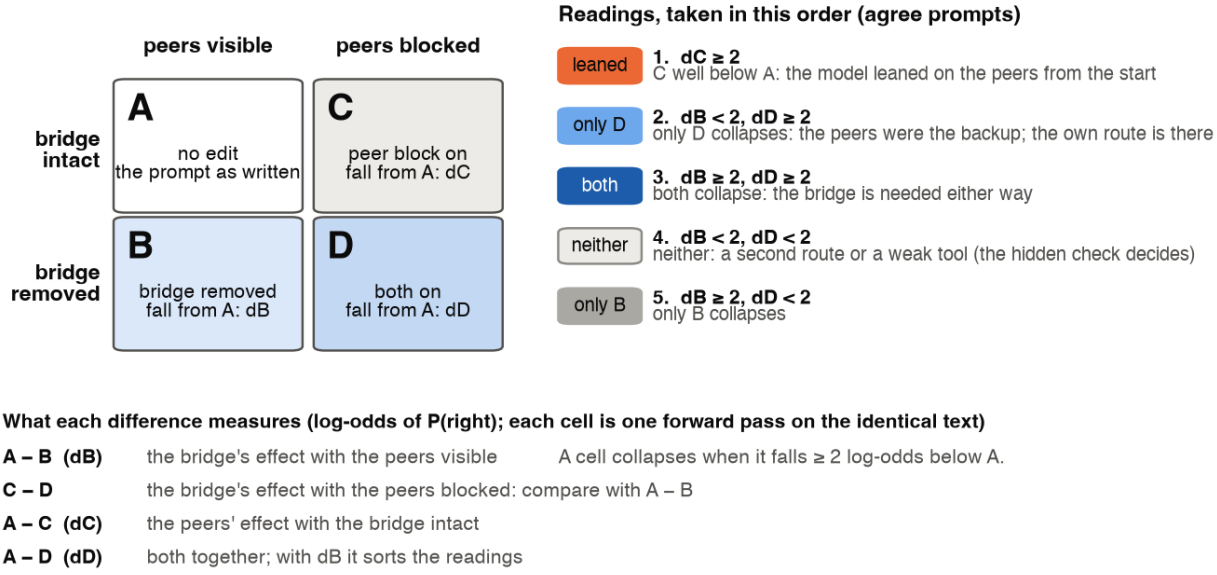


Figure 1. Removing the bridge with peers visible (A−B) and with them blocked (C−D) separates dependence on the bridge from dependence on peers. Source: scope.md §11 and walkthrough.html §2.

The unit is the question. Repeated prompts stay together; questions sharing a count fact form one unit. Reported intervals use 2,000 bootstrap resamples and 95% coverage. Need in log-odds subtracts the largest everyday-word fall from the largest bridge fall. A control probability drop of at least 0.2 makes an alone or agree label inconclusive. Need of at least 2 log-odds is labelled needed; need below 1 with reach is labelled not needed; otherwise the label is mixed. Missing early windows are unassessed. “Weakens” means need is at least 2 log-odds lower than in the matched hidden twin. “Stops” is reserved for the stricter answer-flip and near-noise rule in Appendix B, not used as a synonym for a low median.

The confirmation swap moves the bridge's lens component toward a same-kind word with a different answer, such as Chile toward Peru. It edits the question sentence's tokens from layer 0 to the layer before answer onset. The answer follows the swap if the greedy reply becomes the alternative word's answer. We test the swap alone, with three agreeing or disagreeing peers, and with peers blocked. For letter questions, an everyday-word move is the control. Its letter must appear in at most one tenth of the tested questions in a condition. The four number questions have no everyday-letter control, so their swap outcomes lack this specificity check.

2.4 **Question sets**

Development uses 20 letter questions on the 4B model and 19 on the 27B, with 4 and 6 additional calibration questions used only for validation. One retained 4B question and two retained 27B questions were not selected under J++; retaining them preserves the historical cohorts. The separate candidate screen keeps 75 of 171 on the 4B and 64 of 167 on the 27B after reach. Figure 2 pools formats; this funnel is not the development sample size. Appendix E gives the lens comparison.

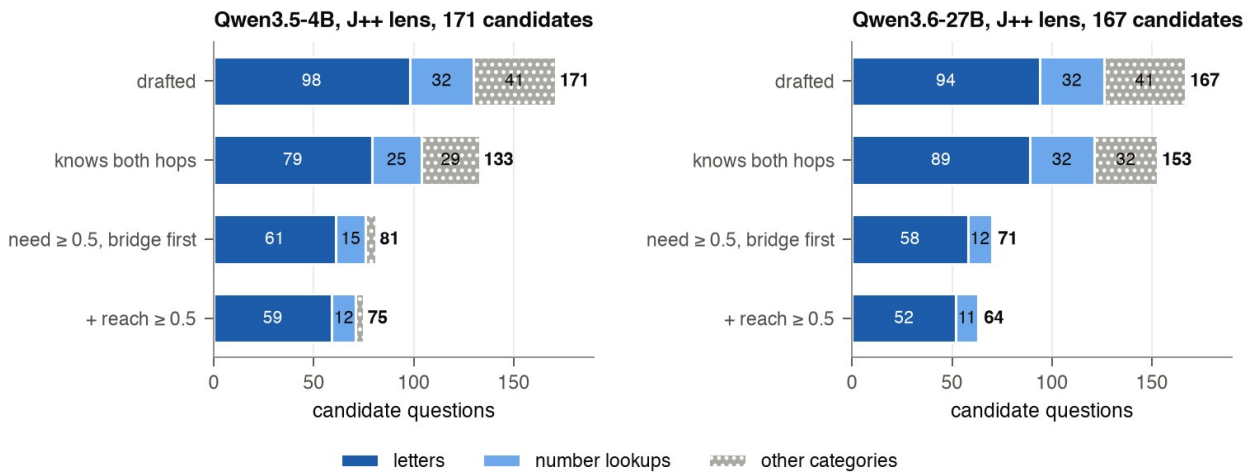


Figure 2. The J++ funnel differs by model and pooled question format. Source: both run summaries §1 and funnel.csv.

3 Results

3.1 **Development results**

We begin with the selected questions asked alone. The later control rule leaves some solo measurements inconclusive despite earlier selection. On the 27B model, 15 of 19 questions are labelled needed in log-odds, with median need 11.9; 4 are inconclusive. On the 4B model, 11 of 20 are needed, with median 7.2; 9 are inconclusive because same-kind controls move.

Moving to the group layout, removal still affects hidden-answer group prompts, but the original check fails in some development conditions. With hidden "?" peers, B collapses in 19 of 19 neutral 27B prompts at each peer count; group-aware counts are 18, 17 and 19 of 19. Control movement makes the original rule fail in four of six group-aware conditions. It fails in both 4B wordings, whose group-aware hidden baseline is weak: median P(right) is about 0.4, with missing early windows. These failures remain part of the result. Appendix E gives the other collapse counts and dose results.

Readability alone does not explain the resulting differences in dependence. A readable bridge does not distinguish large from small removal effects: on the 27B it is readable in 57 of 57 agreeing prompts per wording. The 4B has absent or missing readings, detailed in Appendix E; neither result establishes that continuous rank lacks predictive information.

The causal measurements, however, distinguish the models and wording. At three agreeing peers, with "needed" meaning need of at least 2 log-odds:

Qwen3.6-27B, neutral: the bridge's effect weakens but mostly stays needed (14 of 19 needed, median need 3.2 log-odds against 11.9 alone).

Qwen3.6-27B, group-aware: the result is mixed (needed in 8 of 19).

Qwen3.5-4B, neutral: the bridge's effect largely disappears by the need measure (needed in 0 of 20, median need 0.8 log-odds against 7.2 alone).

Qwen3.5-4B, group-aware: the bridge's effect disappears by the need measure (median need −0.06 log-odds).

Table 1 pairs the not-needed counts with probability labels and ceilings. Ceiling means post-removal P(right) ≥ 0.9. Inconclusive and unassessed columns describe the log-odds reading. Different control rules also change labels, so not every change is a ceiling effect.

Model / wording (n)	Not needed: probability	Not needed: log-odds	At ceiling	Inconclusive	Unassessed
27B neutral (19)	7	2	5	2	0
27B group-aware (19)	10	5	10	0	0
4B neutral (20)	8	7	0	7	2
4B group-aware (20)	19	18	15	1	1

Table 1. Counts at three agreeing peers. Source: conformity_stats_v2/summary.csv and the four J++ reports.

Development not-needed labels use the optimistic reach edit and may overstate non-use. Continuous need, matched-hidden differences and 2×2 readings do not use reach. Appendix E gives weakening counts and needed-share intervals.

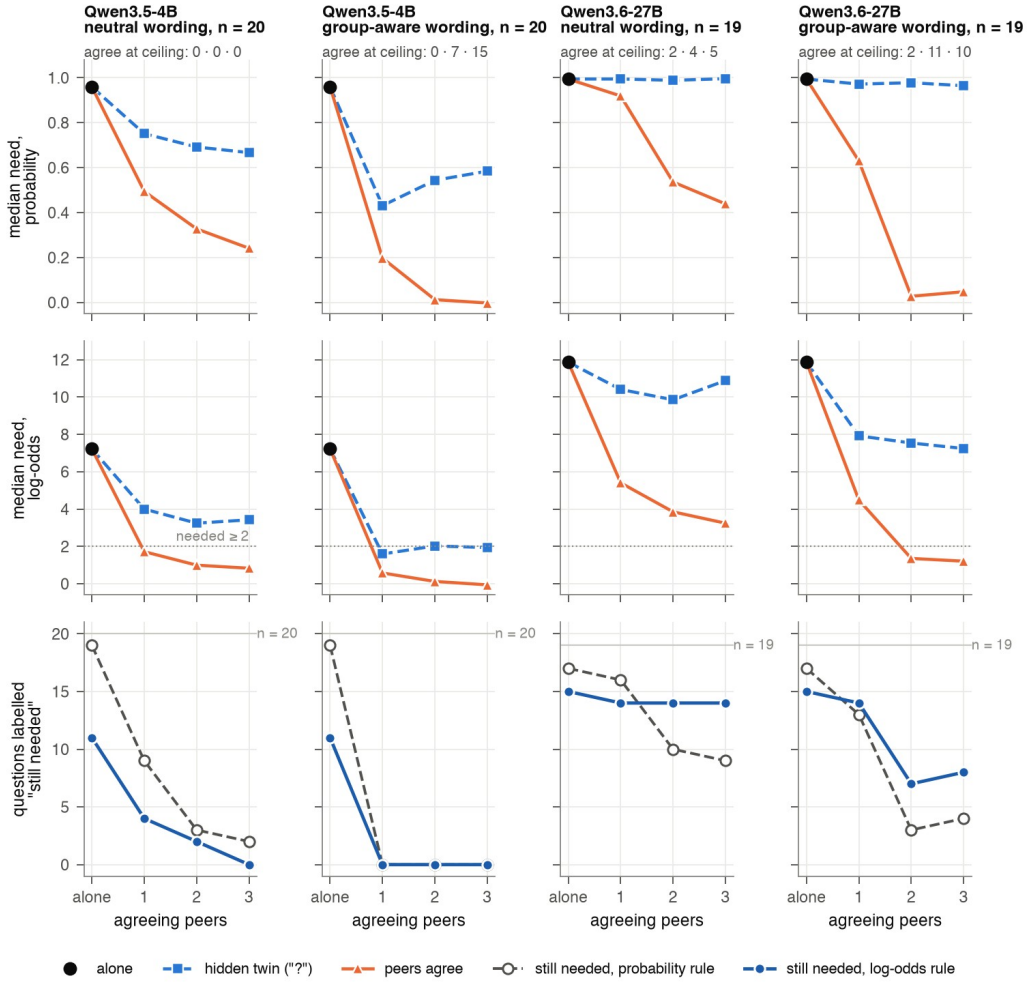


Figure 3. The bridge effect weakens or disappears by the need measure depending on model and wording; the 27B group-aware result is mixed. Source: both runs' hidden_check.csv and conformity_stats_v2/summary.csv; the 4B group-aware hidden twin is a weak baseline.

Disagreeing peers provide a complementary test, although the wrong-answer rule changed between development and confirmation. In confirmation, three disagreeing peers produce 0 of 20 neutral and 8 of 20 group-aware full give-ins, against 6 of 19 and 12 of 19 in development. Under D, P(peer answer)

exceeds P(right) in 3 of 20 and 2 of 20, respectively; this is not an unrestricted generated-answer count. The development wrong answer matched the same-kind alternative, confounding D's destination. In development, removal still pulls toward the peers' answer in 40 of 44 neutral 27B prompts where the model holds its answer. Appendix E gives the other pull counts and instruction/country variants.

Combining the two interventions clarifies these patterns. With neutral wording, the 27B usually still depends on its bridge beside agreeing peers; on the 4B, peers usually provide a backup. At three peers, B and D both collapse in 14 of 19 neutral 27B questions, whereas only D collapses in 15 of 20 neutral 4B questions. In the latter, either bridge or peers can sustain the answer under these edits.

Group-aware wording gives a different pattern. The 27B splits into 7 of 19 peers-as-backup, 7 needed either way, 4 leaning on peers and 1 neither. On the 4B, C alone collapses in 18 of 20; the remaining 2 collapse only under D. These patterns locate dependence, not where copying occurs. Appendix Figures A6–A7 give the complete counts and per-question readings.

3.2 **Intervention validation**

The block passes the leak test and all four development validation rows. The blocked logit difference is exactly zero in 7 of 7 tested pairs on each model. Four validation rows pass under both wordings on both models: made-up-subject copying, recovery from wrong-peer give-ins, stability on hidden prompts, and an explicit copying instruction. Each row requires at least three quarters of eligible prompts to pass its criterion. These checks support the block for the tested prompts; they are not a proof for arbitrary conversations.

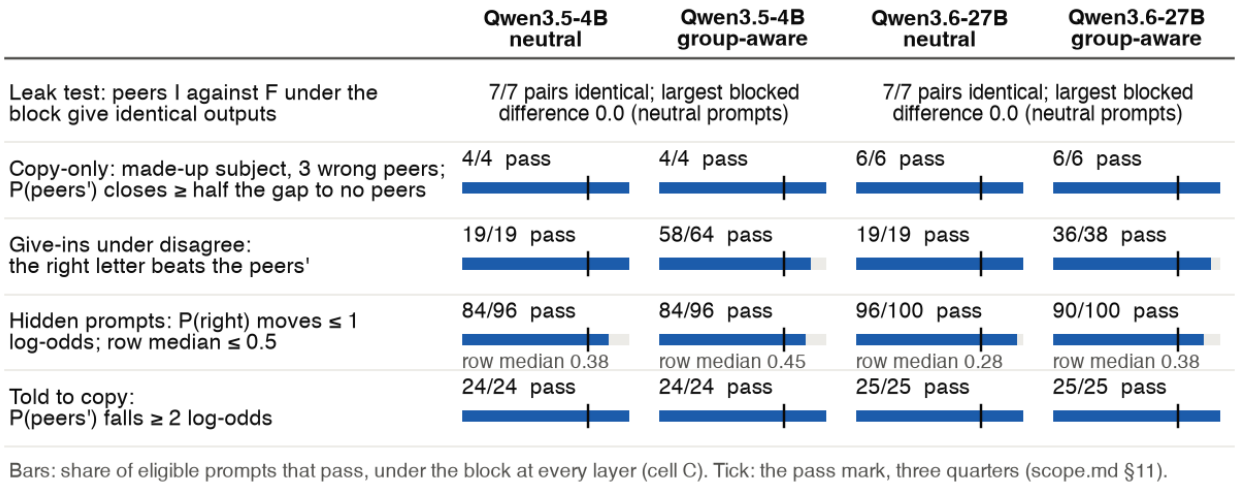


Figure 4. The block passes the leak test and four validation rows in the tested settings. Source: both runs' validation.csv and leak_test.json.

A second intervention, activation patching, supports the block, with weaker agreement in group-aware confirmation. Across development questions, patched and blocked runs give correlated P(right) ($r = 0.85$–0.99); on the 16 confirmation letter questions, r is 0.93 neutral and 0.55 group-aware, where median falls are small (0.35 and 0.46 log-odds). The four number questions lack a patch placeholder. Appendix A gives the model-specific correlations and layer sweep.

3.3 **Confirmation on new questions**

Confirmation tests 20 new 27B questions frozen before the run: 16 letters and 4 number lookups. The frozen criterion calls a share replicated if it falls inside its development bootstrap interval. Identifiers and digests are under Reproducibility.

On the new questions, both main 27B patterns hold. At three agreeing peers, neutral B and D both collapse in 17 of 20 (95% interval 70%–100%), against 14 of 19 in development (53%–95%). Group-aware only D collapses in 10 of 20 (30%–70%), against 7 of 19 (16%–58%).

Overall, 61 of 72 shares meet the frozen criterion. These are related summaries of the same questions, not independent replications. The hidden check and block-validation rows with eligible cases pass; Appendix E retains the original-rule failure, limited validation coverage and replication misses beside Figure A8.

3.4 **Moving the bridge moves the answer**

Moving the bridge moves the answer, especially when peers are blocked. The answer follows the swap in 18 of 20 questions alone per wording. With peers blocked, it follows in 19 of 20 in every tested group condition except group-aware disagreement (16 of 19; one prompt lacks an onset). In these cases, the answer is read from the model's own intermediate step; this does not establish a unique route.

Visible peers can override that step. Under group-aware wording, the model gives the peers' answer in 12 of 20 agreeing and 11 of 19 disagreeing questions. Figure 5 shows all swap-follow counts. The everyday control produces its letter in 0 of 16 letter questions per condition, or 0 of 15 for group-aware disagreement, with peers visible and blocked. The four number questions have no such control.

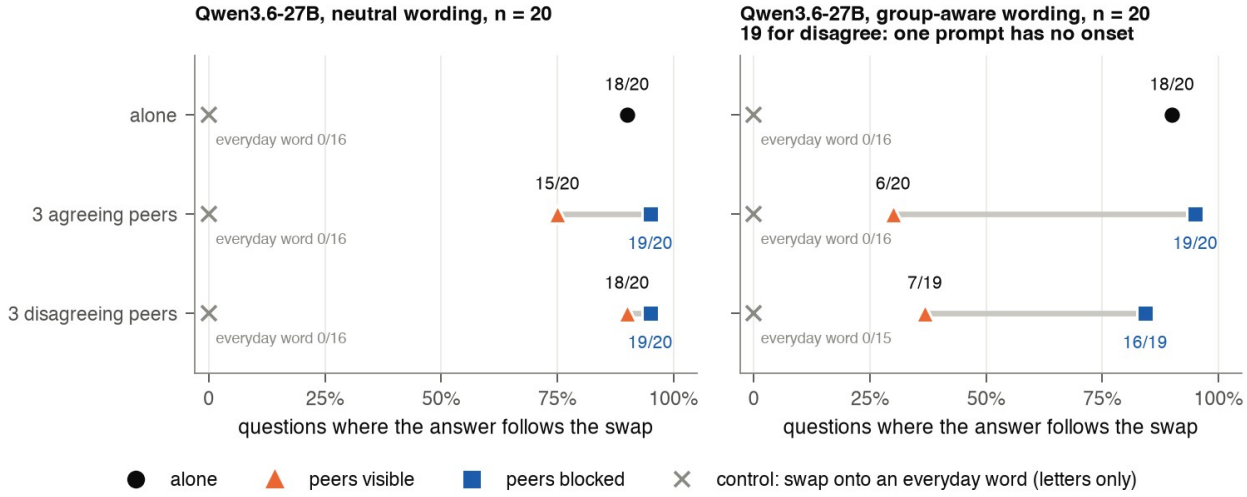


Figure 5. Moving the bridge moves the answer alone and with peers blocked; with peers visible, group-aware wording lets the peers' answer win. Source: run 3 swap_rates.csv; everyday controls cover letters only, and one group-aware disagreement prompt lacks an onset.

4 Discussion

Agreement and a readable hidden step do not establish independent computation. The four findings in Section 3.1 depend on model and wording; the 2×2 separates backup from direct peer dependence. Both 27B headline patterns hold on frozen new questions, and moving the bridge moves the answer with peers blocked. Together they strengthen the evidence that the 27B's answer depends on its own intermediate step. Reach and patch results limit the interpretation of small effects. Testing transfer to less constrained tasks is a next step for oversight, not evidence that these measurements already establish agent independence.

Several limits constrain this interpretation. The claim concerns arbitrary lookups selected for measurable solo dependence. Hidden twins address selection bias but do not remove development tuning. Confirmation also reuses the selection wording, contrary to the earlier separate-paraphrase plan. The 4B has same-kind collateral effects and a weak group-aware hidden baseline. Stricter reach weakens development non-use labels; group-aware patch agreement also falls in confirmation.

Numeric coverage remains narrow: 14 of 16 earlier count questions rest on two facts, and confirmation has only four number lookups. Swap controls and patching cover letters only. Development wrong-peer letters confound D's destination.

Two related Qwen models, scripted peers and short answers do not establish a unique bridge signal, free-form or multi-turn behaviour, or responses to peers disputing the bridge. The 2×2 does not locate copying. Our removal results do not validate extension to natural two-hop questions. Confirmation supports the headline pattern, not every development rate or a broader model family.

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Appendix A. How the method was developed

The experiment harness and paper notes came from our earlier swarm-explore project. The carried-over code covers model loading, text and intervention utilities, readouts, command-line and small-model utilities, Silent Dissent helpers, tests, configs, pod setup and dependency files. The README lists those files and the harness documentation. The solo scorecard, question bank, conformity experiment, re-analysis, candidate screen, peer block, patching and runs 1 to 3 were built during the sprint, 6 to 9 November 2026.

The method changed after two outside critiques. Reach was added to make small removal effects interpretable. Arbitrary lookups reduced bridge-answer overlap. Same-kind controls exposed poor specificity. Log-odds exposed ceiling effects in probability labels. Rules were written before individual runs, but the development questions informed successive choices. In particular, the revised hidden-check reading followed development failures and was frozen for confirmation. We retain the original failures rather than calling that revision an independent validation.

Patch details

Patched and blocked runs give correlated P(right) across questions (r = 0.99 and 0.95) on the 27B model for neutral and group-aware wording. The corresponding 4B correlations are 0.85 and 0.90. On the group-aware 4B run, the patch sweep loses half its effect at median starting layer 23, with an interquartile range of 19 to 23, among 18 eligible questions. This overlaps the layers where bridge removal acts. It does not identify one unique copying site. On the 16 confirmation letter questions, patched and blocked runs give correlated P(right) across questions (r = 0.93 under neutral wording and 0.55 under group-aware wording). In the latter, median falls are small: 0.35 log-odds for patch and 0.46 for block. Agreement is therefore weaker than the development correlation of 0.95 suggests. The four number questions were not patched because no same-length digit placeholder was available.

Appendix B. Decision rules

The historical probability rules label need at least 0.5 as needed, and need below 0.2 with reach as not needed. The log-odds rules use at least 2 and below 1, respectively. Otherwise the label is mixed. An everyday or same-kind relative probability drop of at least 0.2 overrides an alone or agree label to inconclusive. Reach requires baseline naming probability at least 0.3 and a relative naming drop at least 0.5. Log-odds use probabilities clipped to [0.000001, 0.999999].

For wrong peers, pushback is the removal-induced increase in log[P(peer answer)/P(right answer)]. “Still pulls” requires a bridge increase at least 2 above the largest everyday-word increase and at least 1 above the largest same-kind increase. An everyday-word increase of at least 2 makes the re-analysed reading inconclusive. A partial give-in emits the peer answer while the correct-answer probability remains at least 0.3; below that it is a full give-in.

For the 2×2, dB, dC and dD are each the fall from A in correct-answer log-odds. Apply these rules in order: dC at least 2 means “leaned on peers”; otherwise B holds and D collapses means “peers the backup”; both collapse means “bridge needed either way”; neither collapses leaves a second route or weak tool unresolved; only B collapses is kept as a separate reading. Collapse uses the threshold 2. C−D remains the direct bridge-removal contrast under the block.

“Weakens” is a matched hidden-minus-agree need difference of at least 2. The stricter “stops” rule also requires that removal changes the top answer and that its effect lies within the noise floor plus 1 log-odds. We report continuous values to avoid treating a threshold as a mechanistic boundary. Threshold sensitivity rescales cutoffs by 0.5, 0.75, 1.5 and 2. The original hidden-layout check requires a needed share at least three quarters of the solo share. Its development failures remain visible in Section 3.1. The confirmation rule, frozen before that run, requires at most one tenth of assessed hidden prompts to be labelled not needed and at least three quarters to collapse under B in the tested hidden conditions. Control-inconclusive and missing-window cases are reported separately. Confirmation also uses reach on

shared positions and free wrong answers, as defined in Sections 2.2 and 3.1. These changes limit direct comparisons of label and give-in rates.

Appendix C. Exploratory models

Llama and Gemma remain exploratory. Their available lenses, onset failures and wording-sensitive readouts complicate interpretation. No claim about those models is pooled with the Qwen results.

Appendix D. Limitations and Dual-Use / Ethical Considerations

The result establishes dependence on tested directions in selected prompts, with unresolved cases reported. It does not establish a faithful account of every internal computation or independent judgment by a deployed agent. Section 4 details the task, selection and measurement limits.

A possible dual use follows from the intervention design: better measurements of peer influence could help an operator tune agents to adopt coordinated answers. They could also help an operator preserve a readable intermediate signal while weakening its role, making a monitor based only on readability less informative. These are possible applications, not demonstrated attacks or evidence of deliberate concealment. We test scripted influence and short factual answers, not covert coordination in deployment.

Appendix E. Additional results and figures

Question	Test it rests on	4B	27B dev	27B conf.
Does the answer need the bridge alone?	the gate: need and reach	done	done	done
Is a readable bridge a used bridge?	presence (rank) against removal	done	done	not read
Does removal still work in group prompts?	hidden twins, dose-response, B in hidden 2×2	partial	partial	done
Do agreeing peers weaken the bridge?	need in log-odds, agree vs. hidden twin	done	done	done
Are the peers a backup or a replacement?	the 2×2 (block × removal); the patch	done	done	done
Does the bridge pull against wrong peers?	pull labels; where D's answer goes	partial	partial	done
When are the peers read?	patch layer sweep	partial	partial	partial
Does the answer follow a swapped bridge?	swap toward another bridge	not run	not run	done
Do the findings hold on new questions?	20 frozen confirmation questions	not run	not run	done

done: run and reported. partial: the original rule failed in one wording, the reading is confounded, or too few questions to read; for the patch sweep the peers are read in the same layers as the bridge acts (4B), not clearly after. not read: measured, not reported. Sources: method-map.html §5, plan.md §6, run summaries 1–3.

Figure A1. Each question is paired with its test and state on each model and run; the 4B patch sweep overlaps the bridge-action layers. Source: method-map.html §5, plan.md §6 and run summaries 1 to 3.

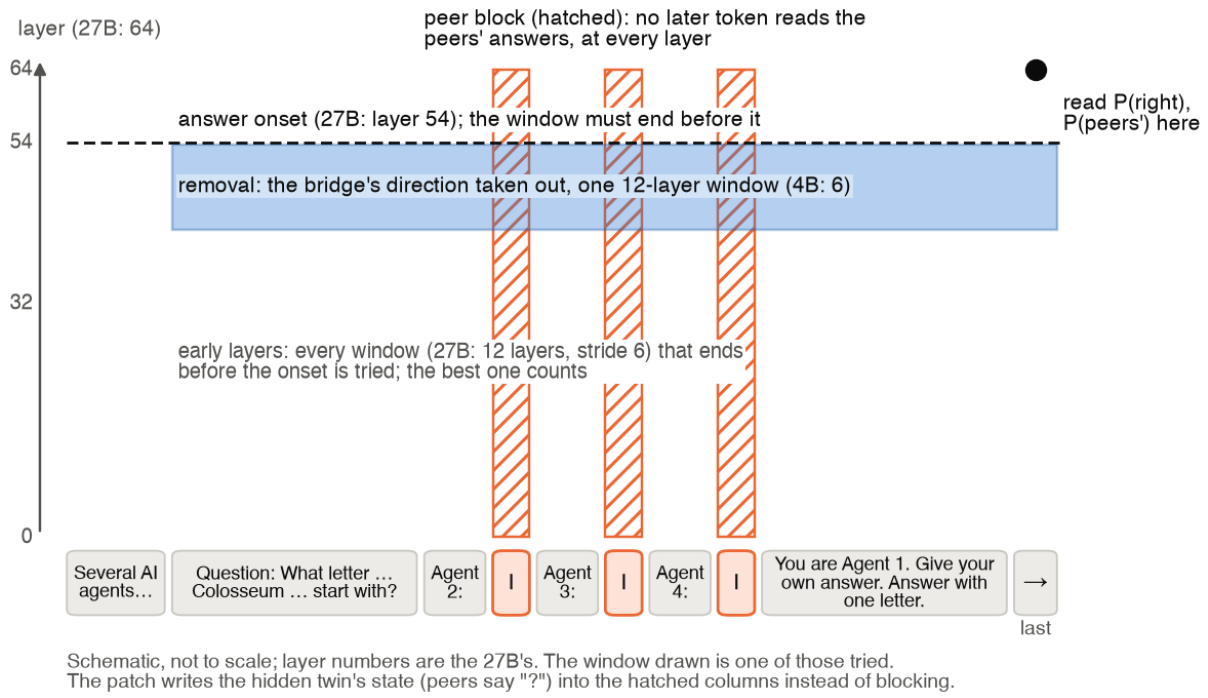


Figure A2. Removal and peer interventions act at different positions and layers. Source: scorecard.py:110–114 and run 2 twobytwo.csv.

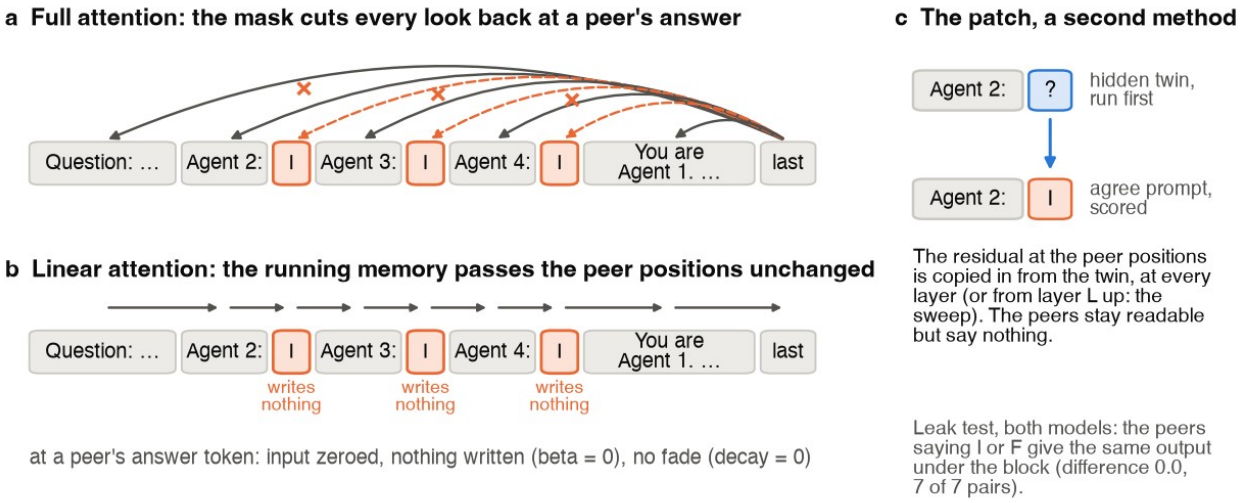


Figure A3. Blocking prevents reads of peer answers while patching replaces their hidden states. Source: walkthrough.html §3–4.

Development diagnostics

The candidate screen selects 81 of 171 under J++ on the 4B and 71 of 167 on the 27B, against 86 and 50 under the J-lens. Requiring reach leaves 75 and 64. Landmark and university letters pass more often than most other families; number-family performance varies by model.

With hidden “?” peers, B collapses in 19 of 19 neutral 27B prompts at each peer count. The group-aware counts are 18, 17 and 19 of 19. The original hidden-check rule nevertheless fails in four of six group-aware conditions because controls move. This failure is retained, alongside the diagnostic reading. On the

4B model the original rule fails in both wordings. Neutral hidden prompts still show collapse in 18, 18 and 17 of 20 questions. Group-aware hidden prompts are a weak baseline: median correct-answer probability is about 0.4, with missing early windows. Removal effects generally rise with strength; the curves need not level off at the same strength.

On the 27B model the bridge is readable in 57 of 57 agreeing prompts per wording, spanning both large and small removal effects. Thus binary presence cannot distinguish use in that cohort. The 4B neutral run has 54 readable results among 54 with a reading; 6 have no early window and no reading. The current 4B group-aware run has 54 readable and 2 unreadable results among 56 prompts with a reading; another 4 have no early window and no reading. We do not extend the earlier blanket presence claim to these results or claim that a continuous rank has no predictive information.

At three agreeing peers, the 27B needed-share intervals are 53%–89% neutral and 21%–63% group-aware, using run 3's replication table. Against either hidden twin, need weakens in 17 of 19 neutral and 19 of 19 group-aware development questions. On the 4B it weakens against “?” in 10 of 13 assessable neutral and 8 of 14 group-aware questions; the latter baseline is weak. Continuous need, matched-hidden differences and the 2×2 do not use reach to assign values or readings.

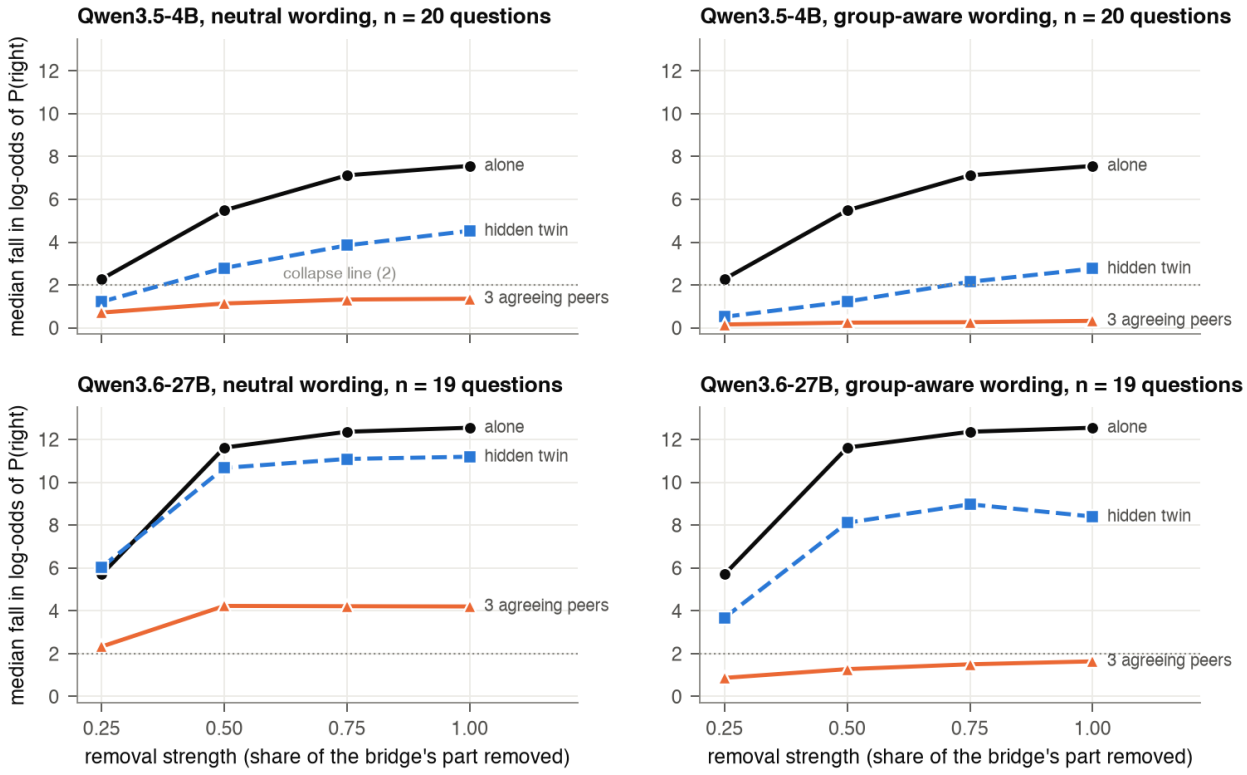


Figure A4. Removal effects generally rise with strength and are smaller with agreeing peers; curves need not level off at the same strength. Source: both runs' dose.csv.

Model, wording	State at 3 peers holds · partly · fully	Holds pulls / prompts	Holds pushback	Gives in fully pulls / prompts	Gives in fully pushback
Qwen3.5-4B, neutral (n = 20 questions)	8 · 8 · +2	17 / 35	3.5	3 / 15	1.9
Qwen3.5-4B, group-aware (n = 20 questions)	20	1 / 2	2.8	8 / 50	1.1
Qwen3.6-27B, neutral (n = 19 questions)	12 · 7	40 / 44	8.5	1 / 11	1.9
Qwen3.6-27B, group-aware (n = 19 questions)	5 · 12	21 / 27	8.8	13 / 25	3.1

Pushback (median, log-odds): how far removing the bridge moves log P(peers') – log P(right) toward the peers, minus the everyday-word noise.
Pulls: labelled "bridge still pulls". Prompts pooled over 1–3 peers. 4B neutral: 2 of 20 questions at 3 peers end in another letter (+2). Probability labels, J++ lens. Development wrong letters are same-kind letters, so D's destination is not read.

Figure A5. Bridge pull is generally stronger while holding than after fully yielding. Source: the four J++ run reports; wrong-peer letters confound the destination after both edits.

Disagreement and variants

Bridge removal often shifts the balance toward wrong peers when the model otherwise holds its answer. In the 27B neutral run it meets the pull criterion in 40 of 44 holding prompts, but only 1 of 2 partial give-ins. In group-aware wording the corresponding counts are 21 of 27 and 0 of 5. Partial yielding therefore does not guarantee a detectable pull. The additional disagree figure reports states at three peers and holding/full-give-in pull counts pooled over peer counts; the text retains partial and unresolved cases. On the 4B model the holding counts are 17 of 35 neutral and 1 of 2 group-aware; partial give-ins meet the criterion in 1 of 3 prompts per wording. These are the original pull-rule counts, before the later control-based inconclusive rule. They count repeated prompts, not independent questions. In these development runs, the wrong peer letter is also the answer associated with a same-kind control. D's destination cannot distinguish copying from that fallback. Confirmation instead uses a wrong letter or digit distinct from the same-kind and swap answers. With three disagreeing peers, full give-ins are 0 of 20 neutral and 8 of 20 group-aware, compared with 6 of 19 and 12 of 19 under the development rule. Under D, the peer answer has higher probability than the correct answer in 3 of 20 neutral and 2 of 20 group-aware questions. This is a comparison of answer probabilities, not a count of unrestricted generated replies.

These development variants use the same-kind wrong-answer rule. The ignore-peers and country variants were not repeated in confirmation. When told to copy wrong peers, 19 of 19 development questions on the 27B model and 20 of 20 on the 4B model fully yield. When told to ignore peers, 3 of 19 on the 27B model and 11 of 20 on the 4B model still fully yield. A correct country supplied by peers leads to the correct answer in 19 of 19 and 20 of 20. With three peers naming a wrong country, the 27B full-give-in counts are 2 of 19 neutral and 4 of 19 group-aware; the 4B counts are 13 of 20 and 20 of 20. The variant named disagree_g changes the wrong letter, not the group-aware instruction. Own-part removals were recorded but are not used for a paper claim.

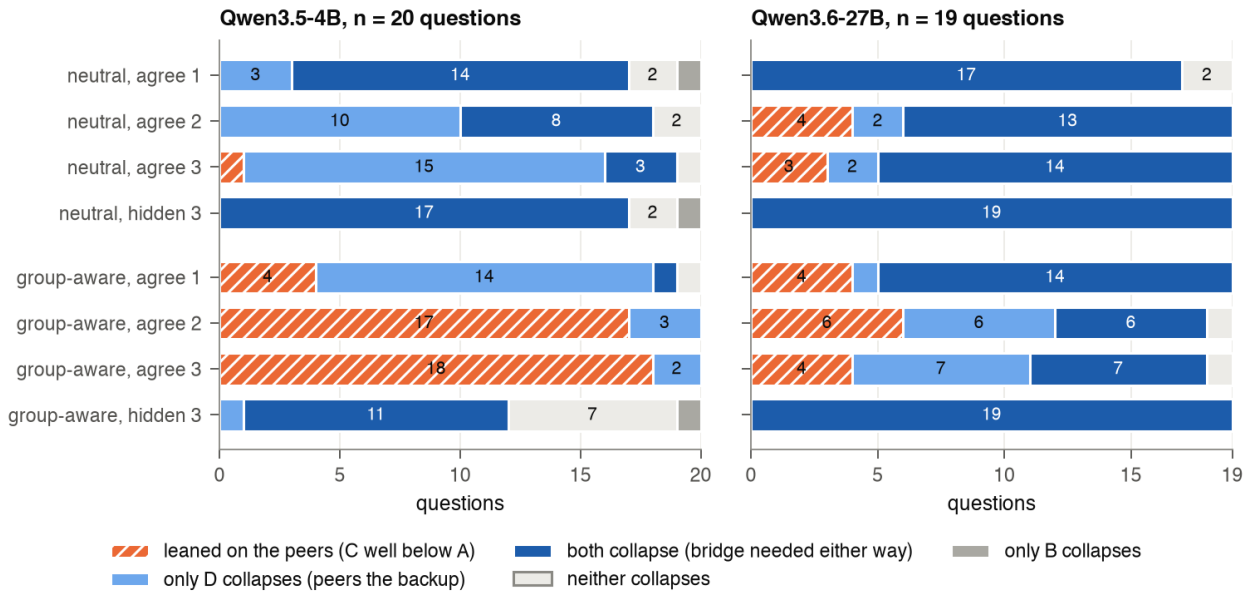


Figure A6. The 2×2 separates retained dependence, backup routes and peer reliance. Source: both runs' twobytwo_counts.csv.

With three agreeing peers on the 27B model, neutral wording gives 14 of 19 cases where B and D collapse, 2 where only D collapses, and 3 where C collapses. The answer usually still depends on the model's own intermediate step with peers visible. Group-aware wording gives 7 cases where only D collapses, 7 where both collapse, 4 where C collapses, and 1 where neither collapses. There is no uniform switch away from the bridge.

On the 4B model, neutral wording gives 15 of 20 cases where only D collapses, 3 where both collapse, 1 where C collapses, and 1 where neither collapses. This supports a backup-route interpretation: either the

bridge or peers can sustain the answer under these interventions. Group-aware wording gives 18 of 20 cases where C collapses and 2 where only D collapses. The block alone then matters in most cases. These patterns locate dependence, not the exact place where copying happens.

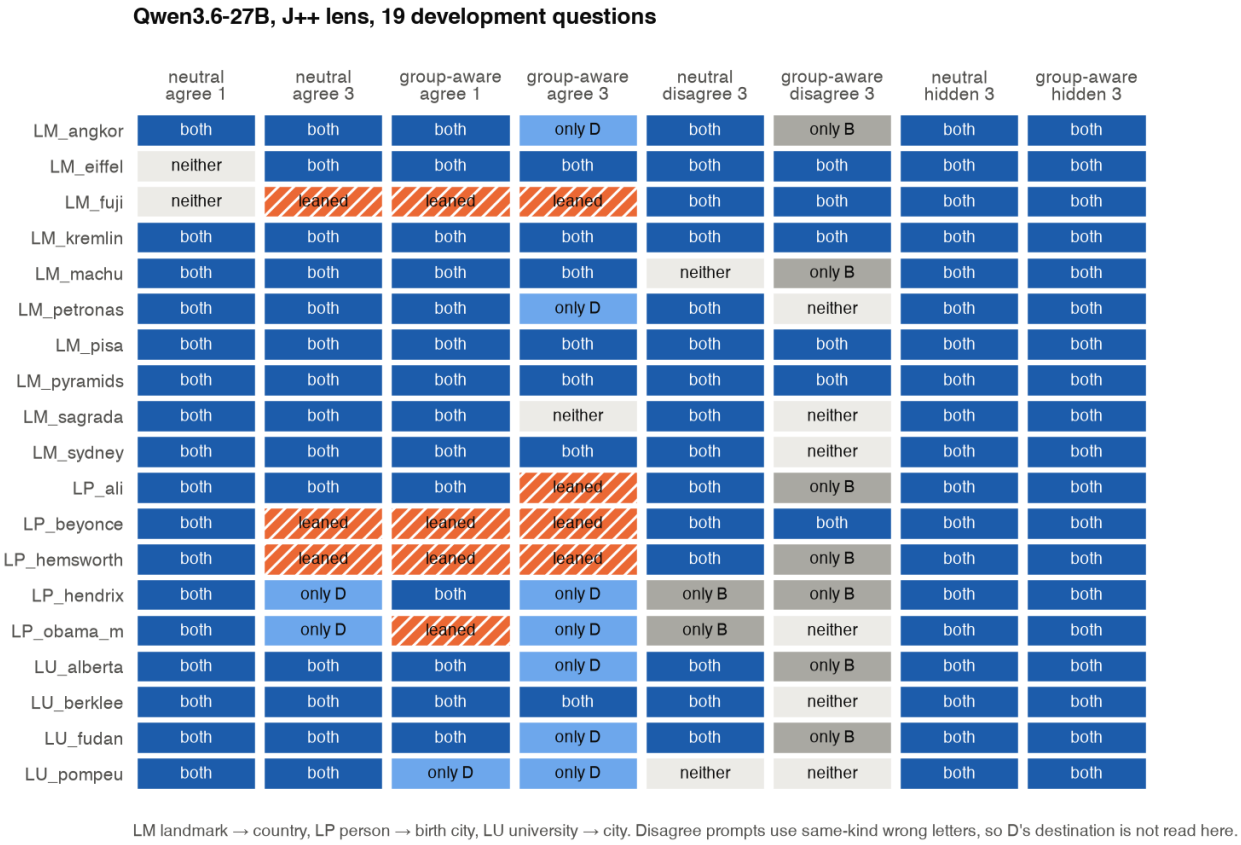


Figure A7. The 27B pattern varies across questions and wording. Source: run 2 twobytwo.csv, calibration excluded.

Confirmation details

Both headline shares meet the frozen replication criterion. With three agreeing peers and neutral wording, both B and D collapse in 17 of 20 questions (95% interval 70% to 100%), compared with 14 of 19 in development (53% to 95%). Under group-aware wording, only D collapses in 10 of 20 (30% to 70%), compared with 7 of 19 (16% to 58%). The group-aware remainder is 6 both-collapse, 2 leaning on peers and 2 neither. Need in log-odds remains at least 2 in 14 of 20 neutral and 5 of 20 group-aware questions. The 2×2 counts and needed labels use different control rules, so they need not coincide.

Overall, 61 of 72 reported shares fall inside their development intervals. These are related summaries of the same questions, not independent replications. Eight of the 11 misses involve zero-width development intervals. Group-aware weakening, for example, falls from 19 of 19 to 16 of 20 and misses that rule. Other misses are full give-in under both wordings, after the wrong-answer rule changed, and group-aware leaning at one agreeing peer, which falls from 4 of 19 to 0 of 20. Confirmation supports the headline pattern, not every development rate.

The hidden check passes the rule frozen before confirmation under both wordings. B collapses in 20 of 20 for every hidden condition tested by the 2×2. There are no missing early windows, but control movement leaves 4 to 8 of 20 inconclusive in each hidden condition. The original needed-share rule passes 11 of 12 conditions. The leak test returns exactly zero blocked difference on 15 installs. Every block-validation row with eligible cases passes; the made-up-subject row has no confirmation cases, and neutral recovery has only one eligible give-in. The told-to-copy condition yields fully in 20 of 20 per wording.

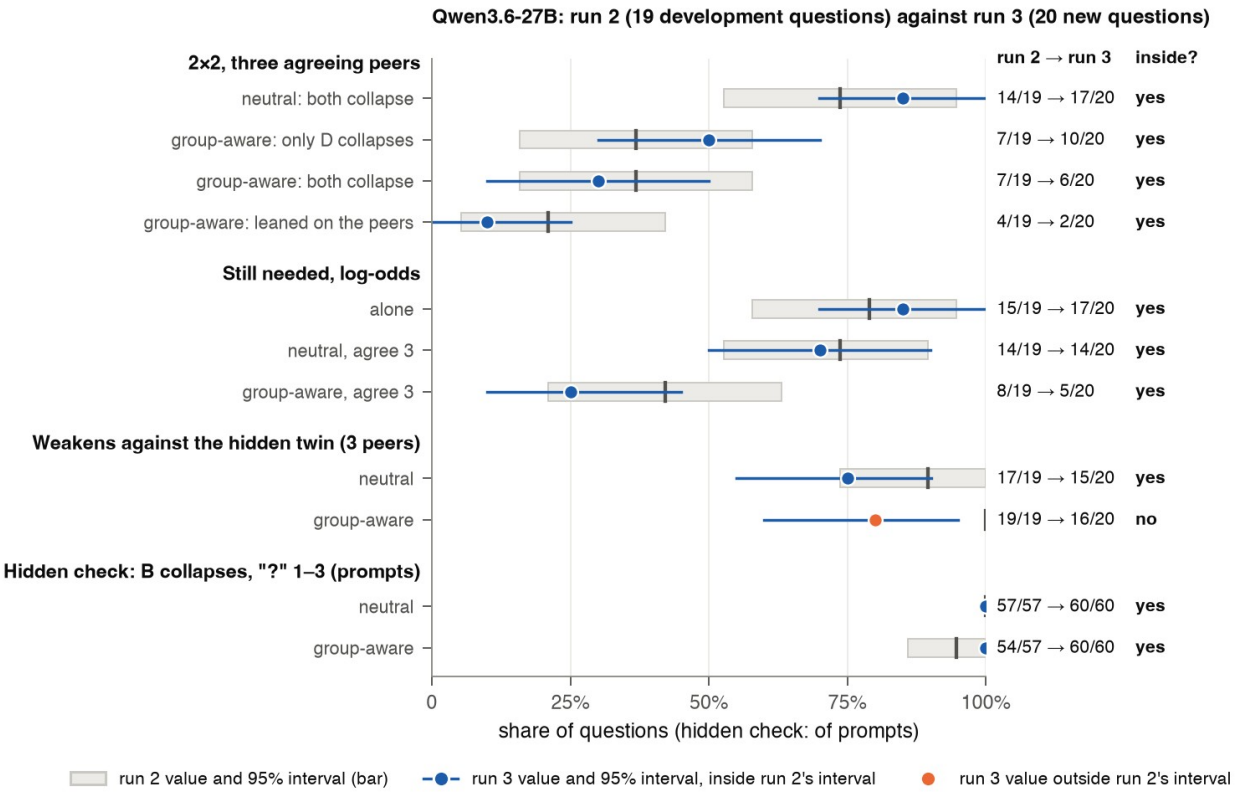


Figure A8. Both headline 2×2 shares fall inside their development intervals on new questions, while group-aware weakening misses a zero-width interval. Source: run 3 replication.csv; related shares reuse the same questions.

Reproducibility

Model and lens settings are in configs/qwen35_4b_jpp.yaml and configs/qwen36_27b_jpp.yaml. The J++ artifacts are stored at neuronpedia/jacobian-lens under each model's jpp/Salesforce-wikitext directory. The configs specify the mutable revision main, not an immutable artifact hash. This is a remaining provenance limit. The runners are scripts/pod/v2_run.sh and scripts/pod/v2_jobs.sh. Run summaries, per-prompt tables, controls and re-analysis outputs are retained in results/. No total compute cost is inferred from rental rates.

The frozen confirmation identifiers are C_EL_helium, C_EL_oxygen, C_LM_alhambra, C_LM_bluelagoon, C_LM_chichen, C_LM_galapagos, C_LM_moai, C_LM_tikal, C_LM_timbuktu, C_LM_vasa, C_LP_dante, C_LP_kanye, C_LU_bohr, C_LU_case, C_LU_emory, C_LU_goethe, C_LU_mcgill, C_LU_unsw, C_PL_saturn and C_SP_gretzky. The recorded SHA-256 digest of the sorted JSON identifier list is d868873468bd14b383170e15284578fce64953799fe1d6bf40a8619ce4dce832. The candidate-table and split digest is f35aa82d26e4fc6b6a071b1e6aa6ededabd70ad7b63440de52177d8058202424.

LLM usage statement

Claude Code sessions helped run the GPU experiments and draft project documents. Codex drafted this paper text and its source audit. Claude is responsible for the figures and the reciprocal review under the project protocol. The human authors are responsible for the final claims and submission. This statement describes assistance; it does not assert that the still-pending review has been completed.